

Sp-23==

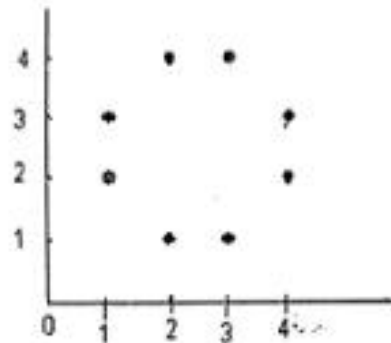
International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Mid Term Examination, Spring 2023
Course Title: Mathematics-IV Course Code: MATH-2407 (New)
Course Title: Mathematics-V Course Code: MATH-3501 (Old)

Total Marks: 30

Time: 1.5 hours

Answer all the questions; Figures in the right hand margin indicate full marks

1. a)



04 CLO1

Determine whether the relation is reflexive, symmetric, anti-symmetric and transitive

b) Let the function $f: \mathbb{R}^* \rightarrow \mathbb{R}^*$ be defined by $f(x) = x^2 + 2x - 4$ Find $f^{-1}(-3)$

01 CLO1

c) Prove De-Morgan's theorem

02 CLO1

$$(A \cup B)' = A' \cap B'$$

d) Evaluate $\int_C (x^2 + jy^2) dz$ from $z = 0$ to $z = 1 + 2j$ along the contour C defined by the line from $z = 0$ to $z = 1 + j$ and then the line from $z = 1 + j$ to $z = 1 + 2j$.

03 CLO2

2. a) A circle in the z -plane has its centre at $z = 3$ and a radius of 2 units. Determine its image in the w -plane when transformation by $w = \frac{1}{z}$. Show your analysis with necessary justification.

06 CLO2

Or

State and Prove Demoivre's theorem

CLO2

b) Prove that $f(z) = |z|^2$ is not harmonic functions but $f(z) = \ln(|z|^2)$ is harmonic.

04 CLO2

3. a) Determine whether $f(z) = z^2 + 5jz + 3 - j$ is analytic

03 CLO2

b) Evaluate $\int_C \frac{z^2 - z + 2}{z - 2} dz$ Where c is the circle i) $|z| = 2$ ii) $|z| = \frac{1}{3}$

04 CLO2

Or

Using Demoivre's theorem find the quadratic equation whose roots are the n th power of the roots of the equation, $z^2 - 2x \cos \theta + 1 = 0$

CLO2

c) Let $f(z) = \frac{2 + z + z^2}{(z - 2)(z - 3)(z - 4)(z - 5)}$ Show all the poles and compute their residues

02 CLO2

d) Draw the graph for the equation $u^2 = -4.5(v - 3)$

01 CLO1



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Mid Term Examination

Program: B. Sc. in CSE
Course Code: MATH-2407
Time: 1:30 hours

Semester: Autumn-2023
Course Title: Mathematics-IV
Total Marks: 30

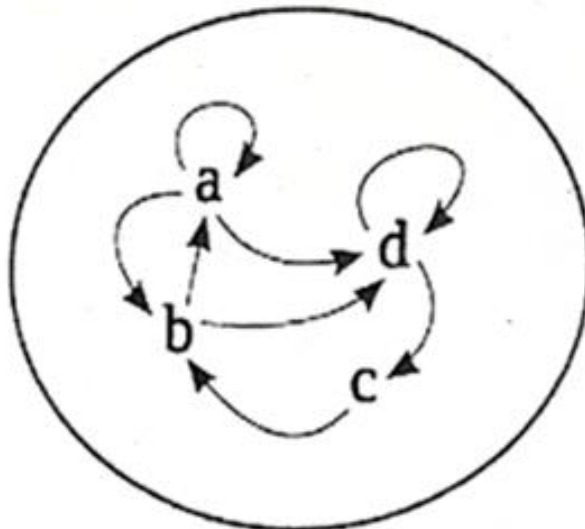
- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
- (ii) Please answer the several parts of a question sequentially.
- (iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

		Marks	CLO	DL
1. a)	Let the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $y = f(x) = x^2 + x - 2$ then find the value of $f^{-1}(-5)$	2	CLO1	C2
b)	If $z = x + iy$ then find the locus represented by $ z - 3 = 4$	3	CLO1	C2
c)	Using De-Moivre's theorem find the quadratic equation whose roots are the n th power of the roots of the equation, $x^2 - 2x \cos \theta + 1 = 0$	5	CLO1	C3

Or



State whether the relation is reflexive, symmetric, transitive and anti symmetric.

2. a) A straight line joining $A(-j)$ and $B(2 + j)$ in the z -plane. Determine its image in the w -plane when it is transformed by $w = \frac{1}{z}$

7

CLO1 C2

- b) Test the function $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic or not.

3

CLO2 C3

Or

Determine the function, $w = e^z$ is regular (analytic) or not.

4

CLO2 C3

3. a) Evaluate the integral $\int_c \bar{z} dz$ from $z = 0$ to $z = 1 + i$ along the curve c .

4

CLO2 C3

- b) Using Cauchy's Integral Formula evaluate $\int_c \frac{z}{z^2 - 3z + 2} dz$ where c is the

circle $|z - 1| = \frac{1}{2}$

2

CLO2 C2

- c) Find the residue of $\frac{1 - 3z}{z(z - 3)(z - 1)}$ at pole $z = 0$

Or

Find the singular points of $w = f(z) = \frac{1}{(z - 1)(z - 4)}$

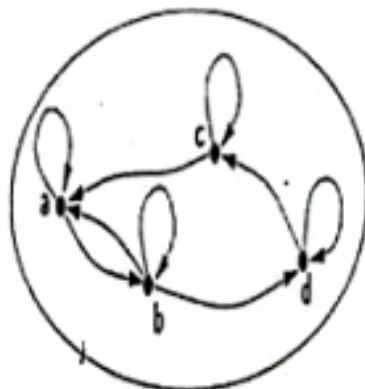
CLO2 C2

Time: 1:30 Hours

Marks: 30

[Answer the following questions. Figures in the right margin indicates full marks]

- | | | Marks | CLO | DL |
|---|---|-------|------|----|
| 1 | a) Let the function $f: R^+ \rightarrow R^+$ be defined by $y = f(x) = x^2 + x - 1$ then find the value of $f^{-1}(10)$ | 2 | CLO1 | C2 |
| | b) Prove De-Morgan's theorem $(A \cup B)^c = A^c \cap B^c$ | 2 | CLO1 | C2 |
| | c) | 2 | CLO1 | C2 |



Determine whether the above relation is Anti-symmetric and Transitive

- | | | | | |
|----|--|---|------|----|
| d) | Using Demoivre's theorem find the quadratic equation whose roots are the n-th power of the roots of the equation, $x^2 - 2x \cos \theta + 1 = 0$ | 4 | CLO1 | C2 |
|----|--|---|------|----|

Or

If $(1 + i\frac{x}{a})(1 + i\frac{x}{b})(1 + i\frac{x}{c}) \dots = A + iB$, Then prove that

4 CLO1 C2

$$(1 + \frac{x^2}{a^2})(1 + \frac{x^2}{b^2})(1 + \frac{x^2}{c^2}) \dots = A^2 + B^2$$

- | | | | | |
|---|--|---|------|----|
| 2 | a) A straight line joining $A(-j)$ and $B(2 + j)$ in the z-plane is mapped onto the w-plane by the transformation equation $w = \frac{1}{z}$ | 8 | CLO1 | C2 |
| | b) Test the function $f(x, y, z) = x^2y + y^2z + z^2y$ is harmonic or not. | 2 | CLO2 | C3 |
| | Or
Determine the function, $w = e^z$ is regular (analytic) or not. | 2 | CLO2 | C3 |

- | | | | | |
|---|---|---|------|----|
| 3 | a) Evaluate the integral $\int_c z dz$ from $z = 0$ to $z = 2 + i$ along the curve c. | 6 | CLO2 | C3 |
|---|---|---|------|----|

- | | | | | |
|----|--|---|------|----|
| b) | Using Cauchy's Integral Formula evaluate $\int_c \frac{z}{z^2 - 3z + 2} dz$ where c is the | 4 | CLO2 | C3 |
|----|--|---|------|----|

circle $|z - 1| = \frac{1}{2}$

Or

- | | | | | |
|--|---|---|------|----|
| | Evaluate $\int_c \frac{2z + 1}{z^2 + z} dz$ Where c is the circle $ z = \frac{1}{2}$ | 4 | CLO2 | C3 |
|--|---|---|------|----|

AU==22

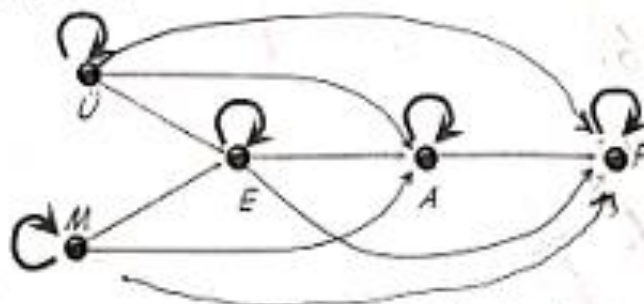
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Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination, Autumn-2022
Course Code: MATH-2407, Course Title: Mathematics-IV

Time: 1:30 Hours

Marks: 30

[N.B. Please answer the questions sequentially. Figures in the right margin indicates full marks]

- | | | | | |
|-------|--|------------|-------------|----------|
| 1. a) | Let the function $f: \mathbb{R}^* \rightarrow \mathbb{R}^*$ be defined by $y = f(x) = x^2 + x - 2$ then find the value of $f^{-1}(10)$ | Marks
3 | CLO
CLO1 | DL
C2 |
| b) | | 3 | CLO1 | C2 |



Determine whether the above relation is Transitive, Anti-symmetric and Transitive Reflexive

- | | | | | |
|-------|---|---|------|----|
| c) | Using De Moivre's theorem find the quadratic equation whose roots are the n th power of the roots of the equation, $x^2 - 2x \cos \theta + 1 = 0$ | 4 | CLO1 | C2 |
| | Or | | | |
| c) | If $(1 + i\frac{x}{a})(1 + i\frac{x}{b})(1 + i\frac{x}{c}) \dots = A + iB$, Then prove that
$(1 + \frac{x^2}{a^2})(1 + \frac{x^2}{b^2})(1 + \frac{x^2}{c^2}) \dots = A^2 + B^2$ | 4 | CLO1 | C2 |
| 2. a) | A circle $ z - 3 = 2$ in the z -plane. Determine its image in the w -plane when transformation by $w = \frac{z}{z+1}$ | 8 | CLO1 | C2 |
| b) | Test the function $f(x, y, z) = x^2y + y^2z + z^2y$ is harmonic or not. | 2 | CLO2 | C2 |
| | Or | | | |
| b) | Determine the function, $w = e^z$ is regular (analytic) or not. | 2 | CLO2 | C2 |
| 3. a) | Evaluate the integral $\int_C z dz$ from $z = 0$ to $z = 1 + i$ along the curve c . | 6 | CLO2 | C2 |
| b) | Using Cauchy's Integral Formula evaluate $\int_C \frac{z}{z^2 - 3z + 2} dz$ where c is the circle $ z - 1 = \frac{1}{2}$ | 4 | CLO2 | C2 |
| | Or | | | |
| b) | Evaluate $\int_C \frac{2z+1}{z^2+z} dz$ Where c is the circle $ z = \frac{1}{2}$ | 4 | CLO2 | |

